



Plant Disease Detection

A project to detect and diagnose plant diseases with the use of machine learning.

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Overview

Agriculture remains the backbone of global food production, contributing significantly to economies worldwide.

Plant diseases are a major cause of reduced agricultural productivity and economic loss, especially in countries where agriculture plays a vital role in the economy.

Our project aims to make the detection and diagnosis of plant diseases easier to access and provide treatment options.



Tech Stack

Frontend

Using a React frontend with TypeScript and Tailwind CSS, we created a user friendly and interactive website for our target audience.

Backend

A python flask backend helps integrate our CNN models with the frontend and creates a pipeline for data processing.

Database

Using supabase, we store the user submitted pictures, along with the predicted disease and the confidence of our models.

Objectives and Goals

1. To provide an automated system for early and accurate detection of plant diseases in potato, tomato, and cauliflower crops using leaf images
2. To reduce dependency on manual inspection and expert knowledge by enabling disease identification through an intelligent computer-based system.
3. To improve decision support for farmers and agricultural practitioners by presenting disease predictions along with confidence levels

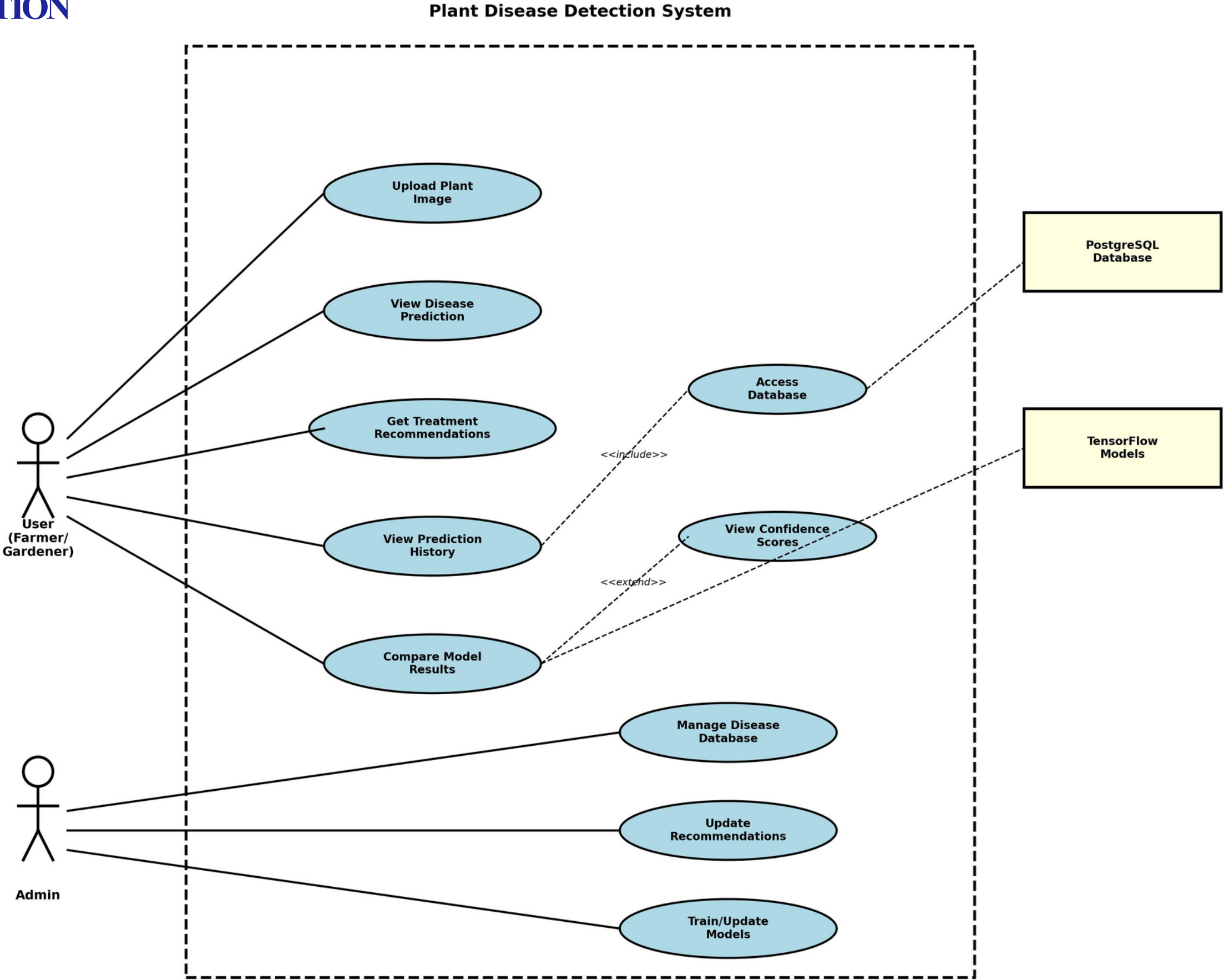


Features & Functionalities

- Automated detection of early and late blight diseases in tomato, potato, and cauliflower plants using leaf images
- Web-based image upload and prediction interface



Use Case Diagram - Plant Disease Detection System



PLANT DISEASE DETECTION

Frontend: TypeScript, Vite, Tailwind
Backend: Flask (Python)
Database: PostgreSQL
ML: TensorFlow, NumPy, Pillow

Plant Disease Detection System - Flowchart

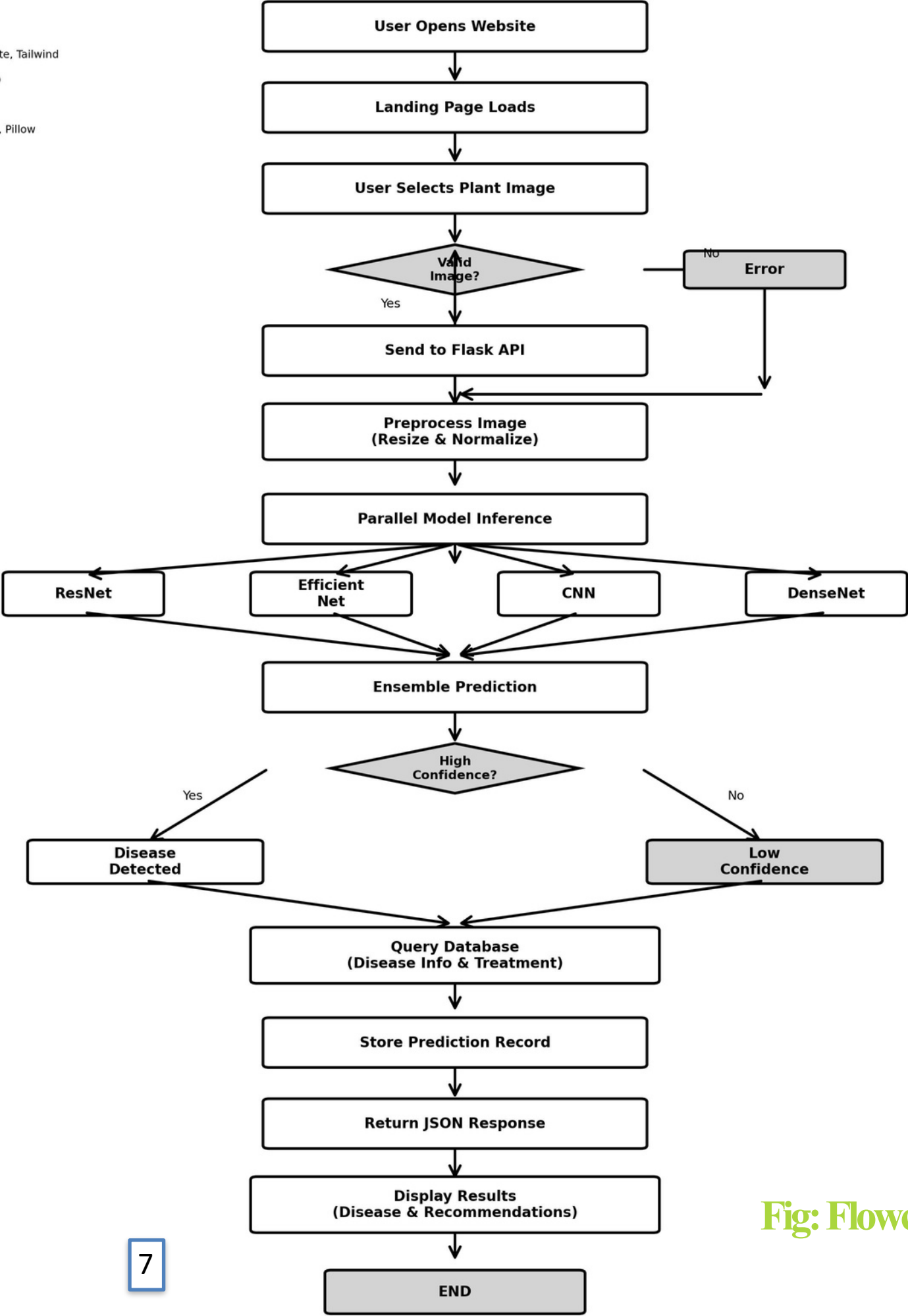


Fig: Flowchart

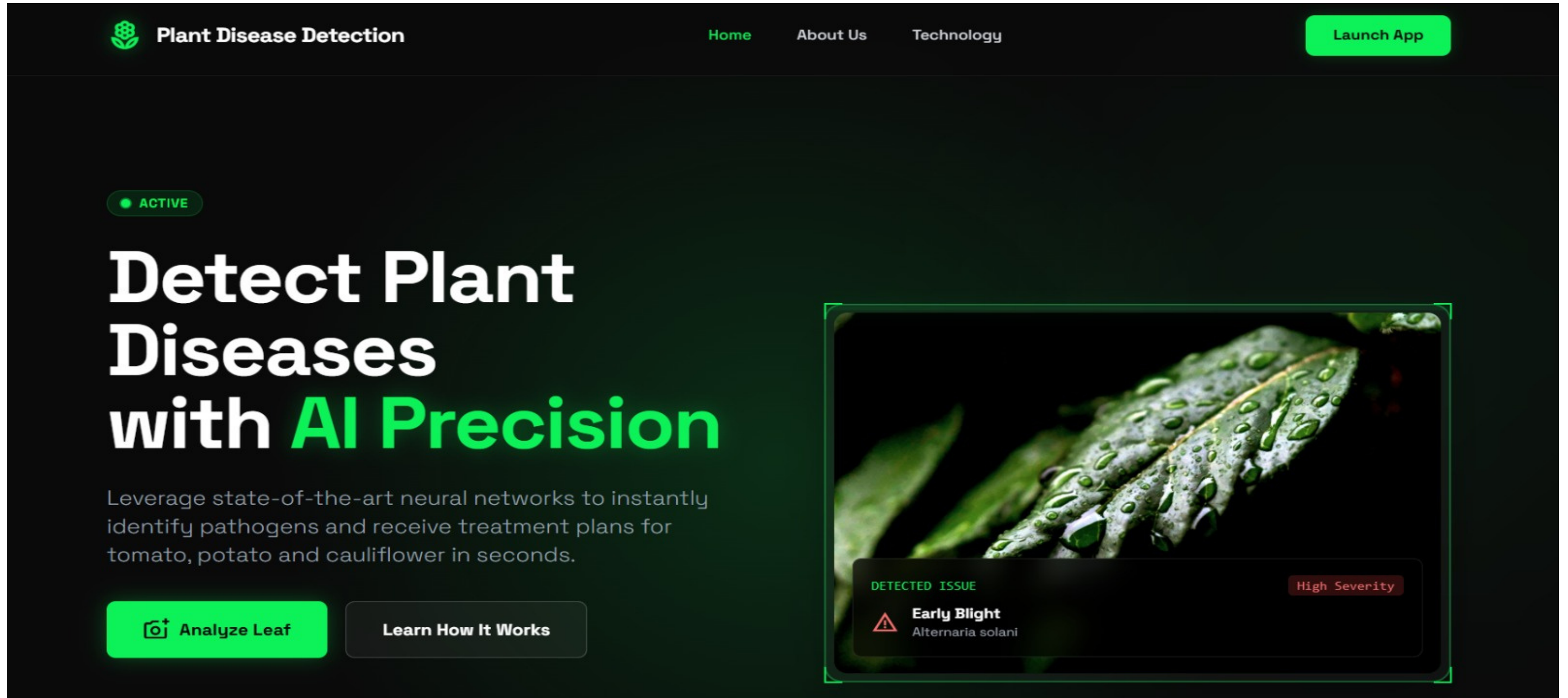


HOW TO USE

Using our website is as easy as 1, 2, 3

Home Page

1) At the home page, press the “Analyse leaf” button



Choose your CNN

2) Upload your image, pick a CNN, and press the button

(You can analyse your image multiple times with different CNNs!!)



Select Leaf Image
Supported formats: JPG, PNG, TIFF (Max 25MB)

 **BROWSE DEVICE**

 **Select Neural Network**

 **ResNet-50**

ACCURACY

98.5%

LATENCY

45ms

 **EfficientNet-B0**

ACCURACY

96.2%

LATENCY

12ms

 **DenseNet-121**

ACCURACY

97.8%

LATENCY

68ms

 **CNN Basic**

ACCURACY

89.4%

LATENCY

15ms

 **ANALYZE IMAGE**

10

Results

3) Get your results and the treatment protocols

 **PLANT DISEASE DETECTION**
ANALYSIS REPORT

NEW SCANRESULTS

 ANALYSIS VERIFIED

Diagnosis Results

Processed in 5.06s

← BACK TO DASHBOARD



PATHOGEN DETECTED

DETECTED DISEASE

Tomato Late Blight



98.14%

CONFIDENCE

 **Treatment Protocol**



EMERGENCY

Strict Isolation

Treatment Protocol



EMERGENCY

Strict Isolation

- Bag and remove whole plant.
- Apply systemic fungicides.
- Alert neighboring growers.



Forestry Department Database

Access regional blight reports & guidelines



Agricultural Research Portal

Latest studies on disease resistance





Limitations & Challenges

- **Limited data size** ~18 thousand images
- **Computational Constraints:** Trained models on personal computers.
- **Controlled Validation Environment:** Validation of model accuracy was done on a controlled dataset
- **Scalability:** The system has only been tested on individual non-concurrent users

Future Enhancements

- **Live Video feed analysis**
- **Larger Dataset with more diverse plants & diseases**
- **Multi-GPU training**
- **Real World Deployment**

Conclusion

With this project, we have built a foundation for easy access to Plant Disease Detection and information on how to treat them. This project successfully trained and evaluated four different Convolutional Neural Network (CNN) architectures for plant disease detection, and a fully functional web-based application was implemented. We hope this technology helps bring stability to the people who need it the most.



Thank You !!